an audio-visual history. It could later be joined with other databases that include more information on national cinemas and television broadcasts. Finally, the content itself, actual films and television programmes are increasingly available in digital format.[36]

Digital technologies of visualization have been, and continue to be, an expanding area of research and development in the 2000s. Visual devices penetrate our present world, from digital cameras and mobile phones to UHD television sets and high-resolution displays. The use of these technologies has taken a central position in how we entertain ourselves, in how we express our relationship with the world around us, and how we are controlled and policed. During recent decades, audio-visual culture has undergone a transformation – from analogue movie theatres to 4k digital cinemas – to such an extent that nitrate and cellulose acetate films need special arrangements and old technology to be able to be shown. Consumers today produce visual and audio-visual content more than ever before, and they share these visualizations on social media and preserve many more on their private hard drives, as well as on cloud services. To be sure, the years to come will bring forth new insights, methods and applications that interpret history both in visual terms and based on visual cultural heritage.

4

Interdisciplinarity: Challenges for Research

Vast resources are available for the digital historian. As the previous discussions show, at the same time as more material is easily available, there are many heuristic and source-critical aspects that have to be taken into consideration. A major challenge is the use of large data sets. Therefore, digital history often appears as interdisciplinary work and takes the form of collaborative research based on cooperation and joint work. There is a need for transgressing disciplinary borders not only because of the complex problems awaiting resolution but also because, in other fields of research, such generic tools have been, or might have been, developed and productively deployed that can be applied in other contexts. Digital historians think about their work in terms of their interest in historical problems, but already problem-solving might require expertise from different disciplines and methodological traditions. It would be unfortunate not to take advantage of what has been found in other scientific communities and within different research paradigms.

Digital history does not stand alone before this challenge. The world has always been complex, to be sure, but we are increasingly conscious of the global hazards we confront, which further emphasizes the need for interdisciplinary work. In his compelling book *We Have Never Been Modern*,

originally published in French in 1991, Bruno Latour started his analysis by describing how a single edition of a newspaper describes the world around us. He observed how chemical and political reactions seemed to intertwine: in the present-day world, human and non-human factors are inseparable, a fact exemplified in Latour's observations on the threat posed by the widening of the hole in the ozone layer. Nature is not something to be externalized from human behaviour and culture.[1] The ozone question aroused fierce public debate in the 1990s, and today we can notice that it was, in the end, possible to change patterns of consumption, influence industry and decision makers and, by these means, avoid the ozone catastrophe. This positive result only confirms the fact that human and non-human agency are linked together.

Today, we have new challenges, such as climate change and global warming, the Earth's diminishing biodiversity and political instability that relates to these larger frameworks. From this perspective it is easy to understand the critique that Western academia has its roots in a world quite different from the present. How can we change universities so that they are simultaneously elastic and resilient to respond to these challenges while remaining firm and determined in their long-term role as cradles of serious research that foster intensive devotion to scholarly problems that widen our knowledge of the world, even in cases where these results have no obvious applicable role in the present day? While it is evident that problems like global warming cannot be solved through the means of one discipline only, it is as apparent that these problems do not mean that disciplinary thinking would be outdated.

Beyond disciplinarity

The interest in overcoming disciplinary borders has certainly been part of science, the arts and humanities for a long time. There is a whole vocabulary to describe the many ways in which disciplines can entangle or intertwine. *Intradisciplinary* suggests a scholarly work that takes place within a single discipline only. It is disciplinary par excellence. *Cross-disciplinary*, in turn, refers to the effort to promote mutual

understanding between disciplines. *Multidisciplinary* work goes further in its aim to bring researchers from different disciplinary backgrounds together to concentrate on a joint agenda so that all members can creatively draw on their specific knowledge and skills. In an *interdisciplinary* setting, the aim is even more ambitious, and researchers attempt to integrate knowledge and methods from various disciplines to create an overarching toolbox for the project at hand. To go even further, *transdisciplinary* research, which is a form of interdisciplinary work, goes a bit further in its dissatisfaction with disciplinary premises. While interdisciplinary thinking still acknowledges the role of disciplines, since it is *inter*, or 'in-between', *transdisciplinary* research tackles the challenge of going beyond disciplinary borders and perspectives to create a larger intellectual framework for research.[2] In her book *Interdisciplining Digital Humanities*, Julie Thompson Klein crystallized the idea in the following way: 'Transdisciplinarity transcends the narrow scope of disciplinary worldviews through an overarching synthesis with new conceptual frameworks or paradigms.'[3]

Based on current research and education under the rubric of 'digital history', it seems fair to argue that digital history follows mainly interdisciplinary and transdisciplinary points of departure. In interdisciplinary cases, disciplinary expertise is necessary for achieving goals, but this endeavour is only possible if disciplines unite forces. Simultaneously, and perhaps more so for the digital humanities than digital history, long-term transdisciplinary strategies are necessary for educating a generation of scholars reared without a disciplinary mindset. Both conclusions are true, since the scope, type and goals of interdisciplinary work vary. We can, no doubt, imagine a future where artificial intelligence has permeated scholarly work to such an extent that data science has become a kind of meta-science and become a substantial presence in any kind of research. In that case, data science diverges from the disciplinary register and must be integrated into researcher training on a general level. This will probably not be the case in the future, as science has always been in a state of entropy: new fields of research emerge constantly. But perhaps we need transdisciplinary centres and hubs to envisage new pathways for research.

In the end, everything depends on how the scope, type and goals of interdisciplinarity have been defined. These three dimensions were itemized by Katri Huutoniemi, Julie Thompson Klein, Henrik Bruun and Janne Hukkinen in their 2010 analysis on how interdisciplinarity was articulated in a selection of concrete research projects.[4] For them, the *scope* refers to the question of what is actually integrated into a particular project in which disciplines meet. Interdisciplinarity in a project is seldom comprehensive but is narrowed down to a few areas. The second dimension is the *type* of interdisciplinarity, which means paying attention to how the research will finally be realized, 'not only in the framing of research problems and coordinating knowledge flows between fields, but also in the execution of research and the formulation and analysis of results'.[5] The third dimension emphasizes the importance of the actual *goals* of the interdisciplinary enterprise. Why does interdisciplinarity take place and how do researchers articulate these goals?[6]

In their article, Huutoniemi et al. maintain that these goals for interdisciplinary work are different in nature: some projects spring out of an interest in gaining more in-depth knowledge, while others have a more instrumental and perhaps non-academic agenda. This is a characteristic of any research, not only digital humanities, but the article makes an important point in acknowledging that these goals and their articulations can be analysed and evaluated. In the case of the digital humanities, however, there is a wider aspect of science policy that should be kept in mind and that goes beyond the question of goals. Most research activities are directed by funding bodies, by research programmes and university strategies, both nationally and internationally. This can be seen as a threat for the freedom of research. What happens if researchers must justify their work through engagement with other fields, particularly in a situation where the funding of more disciplinary-based research is diminishing? Yu-wei Lin expressed this in 2012, highlighting the fact that 'national and international research funding councils have increasingly emphasized that research in humanities should engage with data-intensive and evidence-based academic activities, as those in natural sciences and engineering do'.[7] This concern is understandable, and there might be some

truth in it. Still, it is difficult to find evidence that interdisciplinarity or engagement with data-intensive fields of research automatically translates into a loss for humanities. Interdisciplinarity is never a one-way street. It is a process of mutual exchange in which the humanities' viewpoints may also impact researchers from other disciplines. As the world today is saturated with information technology and digital devices, it is even more urgent that historians and other researchers in the humanities engage with research in such fields as data science and engineering.

Trading across borders

Interdisciplinarity is not a simple counting operation from which the result is only the sum of its parts. In the most fruitful and rewarding cases, it leads to observations that would otherwise not have been possible. Often in research, the findings of one discipline are applicable in another disciplinary context, where they can lead to a breakthrough. It is possible, and constructive, to experiment with methods and research strategies over disciplinary borders. An interesting example is a project in organic chemistry conducted by researchers at the Pedagogical University in Kraków, the Polish Academy of Sciences and the Ulsan National Institute of Science and Technology in South Korea. The point of departure was the major problem of the 'enormous number of different molecules and their fragments that play a role in chemical reactions'. The aim was to try to understand why certain clusters of atoms tend to keep together while repulsing some other groups. The researchers decided to compare chemical units using language tools and split the complex molecules into 'meaningful' substructures. After that, they applied text mining tools, especially topic modelling, in clustering these substructures. This was only an experiment, but in the end, they were able to conclude that semantic information can be 'extracted from chemical corpora using text mining algorithms'.[8] In this case, the tools developed in the digital humanities could be employed in solving problems in natural sciences. It is only a matter of imagination and the courage to experiment with the idea.

Another example of how different fields of research can benefit each other, even fields that at first glance do not seem to have much in common, is the software BLAST (Basic Local Alignment Search Tool), which was developed at the National Center for Biotechnology Information. BLAST finds areas of similarity between biological sequences and can be used to compare, for example, nucleotide and protein sequences. It is a software that all researchers within bioinformatics know. The software itself is, however, generic by nature: it detects similar sequences and can be applied to other kinds of sequences, not necessarily biological. In a recent project at the University of Turku, BLAST was applied in the study of text reuse in old newspapers, which include many errors in optically recognized text, mainly because nineteenth-century papers in Finland used Gothic type, which is difficult for OCR software to process. BLAST is built to be tolerant of errors and mutations and might thus be useful for identifying similarities in messy material. In the project, each character was encoded as an amino acid, and then the whole corpus was run through BLAST. In the end, 61 million occurrences of similarity were detected from a corpus of 5.1 million pages, and these occurrences were clustered into repetition chains.[9] The result was a text reuse database that other researchers can query.[10] This unexpected methodological solution would not have been possible if the research team did not include experts in bioinformatics, but its foundation was the fact that there was an interdisciplinary research group in the first place. Furthermore, it must be added that methods and research questions are always interrelated. It is possible to first identify a problem that needs an interdisciplinary solution, but it is equally important to note that interdisciplinary practices stimulate the team's imagination and help locate new and fruitful research questions, problems that might otherwise have gone unnoticed.

Interdisciplinary work often needs shared spaces. Patrik Svensson has written extensively on the role of spaces in the digital humanities, both concrete locations and abstract spatialities, which is inspiring also for digital history. He has described the digital humanities as a trading zone and a meeting place where people, ideas, tools and methods can mingle.[11] In this sense, the application of BLAST was an

outcome of a practice of 'trading' and 'meeting'. However, cooperation is not always so harmonious. Svensson refers to Mary Louise Pratt's notion of the contact zone. According to Pratt, cultures often meet 'in highly asymmetrical relations of power'. The contact zone can be imbalanced and hierarchical as a platform for communication and does not necessarily lead to trade or mutual exchange.[12] This could happen, for example, in cases when one of the partners does not recognize the premises of the other or if there is no willingness to negotiate on the basic concepts. Max Kemman has pointed out that quite often there is 'knowledge asymmetry in interdisciplinary collaboration'. He refers to a project in which 'historians were unaware how the computer scientists would perform their tasks, and as a result lacked the power to properly negotiate a tool that would work to their satisfaction'.[13] Kemman has discussed this problem further, and in more detail, in his 2019 study *Trading Zones of Digital History*. He based his research on wide-ranging source material, including an online survey, interviews, his own observations and the LDA topic modelling of over 10,000 blog posts by historians between 2008 and 2017. In his conclusion, he noted that there was a feeling of uncertainty in digital history projects, arising on the one hand from historians' hesitation on how to use computational methods, and on the other from data scientists' uncertainty on how their methodological toolbox fits with historical data sets. Kemman's analysis revealed different ways of solving these problems: some historians, which he called 'digital history brokers', were able to use the trading zone as a creative space and negotiate with different groups, including policymakers. On a macro level, however, the solutions of the trading zone had not yet changed research practices.[14]

Towards interdisciplinary practices

The question of uncertainty is critical in any interdisciplinary exploration. One of the reasons for this feeling is without a doubt the tension between disciplinary thinking and interdisciplinary pressure. As Yu-wei Lin pointed out, academics come under significant pressure to break boundaries, and

there is even designated funding for these kinds of projects. Nevertheless, researchers' enthusiasm and commitment to their work can be disciplinary by background: they have come to the field to devote themselves completely to it and have refined their skills specifically from that perspective. Therefore, they might be outside their comfort zone in other areas. Many key concepts in research have a disciplinary linkage, and this in itself has value. We have only to note how profoundly such research fields as cultural and gender studies have influenced humanities since the 1980s and 1990s. It has been fruitful to dedicate oneself to developing, for example, the concept of culture and gender. This is only to say that excellent research can be conducted both from a disciplinary and interdisciplinary point of departure.

Digital history, however, is not a discipline; it is a branch of historical practice that can also be an interdisciplinary enterprise in which the researcher must enter the trading zone. This entrance might raise questions about one's academic identity or at least emphasize the skills and knowledge one has gained over the years. This leads to several questions. Where are the boundaries of one's own skills and knowledge? To what extent is it possible to develop one's own expertise to better match the interdisciplinary challenge, and how does one integrate into broader collaborations, especially if one comes from a scholarly background in which research has traditionally been a solitary enterprise? It is natural to feel uncertain in the face of such questions, but the situation is the same for all involved. One possible strategy for opening doors to interdisciplinary exchange is to try to become more conscious of those transparent, often unarticulated practices that all fields of research contain. Very few historians really explicate how they work in practice with their sources. What is the first procedure they undertake in an archive or when they open a book? The same can be asked of a data scientist. What are the basic practices with a data set? Successful interdisciplinary work must identify what those unarticulated procedures are. These habitual processes have been carried on by tradition and nurtured by academic curricula, and seldom are they encapsulated in a few lines. If we understand how we work, we can better understand where the joint undertaking with big data and other digital resources can lead us.

5

Presenting the Past in the Digital Age

At the heart of digital history, since the early 1990s, there has been an endeavour to utilize the opportunities created by information technology, not only for the study of the past but also for presenting it both in the classroom and before the general public. When Daniel J. Cohen and Roy Rosenzweig published their *Digital History: A Guide to Gathering, Preserving, and Presenting the Past on the Web* in 2006, its introduction stated that the book had emerged in response to the huge expansion of historical content on the World Wide Web: 'Virtually every historical archive, historical society, historic house, and historic site – even the very smallest – has its own website. So does just about every re-enactment group, genealogical society, and body of historical enthusiasts.'[1] The change from the early 1990s to 2006 was really significant: digital resources had multiplied. Simultaneously, the use of information technology by historians had become a matter of course, or at least a dimension that could not be neglected.

This chapter returns to digital historians' interest in using digital tools to present the past. It continues by discussing digital history as public history, then pays attention to the possibilities of visualizing data. The chapter ends with technologies known as augmented and mixed realities and their potential for the digital historian.